

Choose 1 answer:

☐ A The series converges.

☐ B The series diverges.

☒ INCORRECT (SELECTED)
The test is inconclusive.

1 / 2

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{25}{16}$$

Got it, thanks! ^

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1) \left(\frac{5}{4}\right)^{2(n+1)+1}}{n \left(\frac{5}{4}\right)^{2n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1) \left(\frac{5}{4}\right)^{2n+3}}{n \left(\frac{5}{4}\right)^{2n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1) \left(\frac{5}{4}\right)^2}{n} \\ &= \left(\frac{5}{4}\right)^2 \\ &= \frac{25}{16} \end{aligned}$$

2 / 2

Since the limit, $\frac{25}{16}$, is greater than 1, the ratio test tells us that the series diverges.

Related content



Ratio test

Not quite yet...

Try again, [see the answer](#), or [skip for now](#).